

Detecting keyhole pore defects and monitoring process signatures during laser powder bed fusion: a correlation between *in situ* pyrometry and *ex situ* X-ray radiography

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Appending A - Supplementary information:

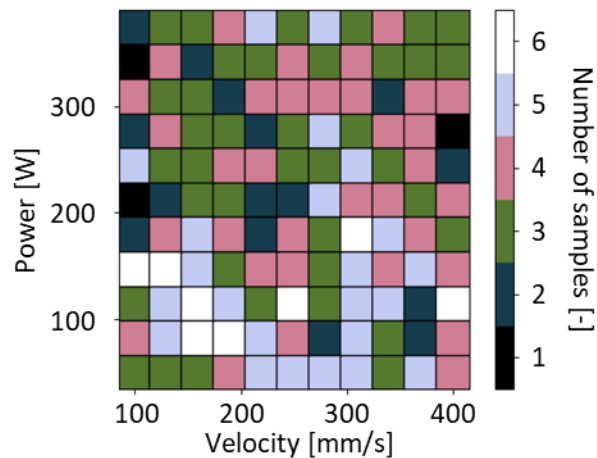


Figure A1: Power-velocity maps of the number of single tracks. Only track that were measured with X-ray radiography were considered in the analysis, resulting in a total number of 446 tracks.

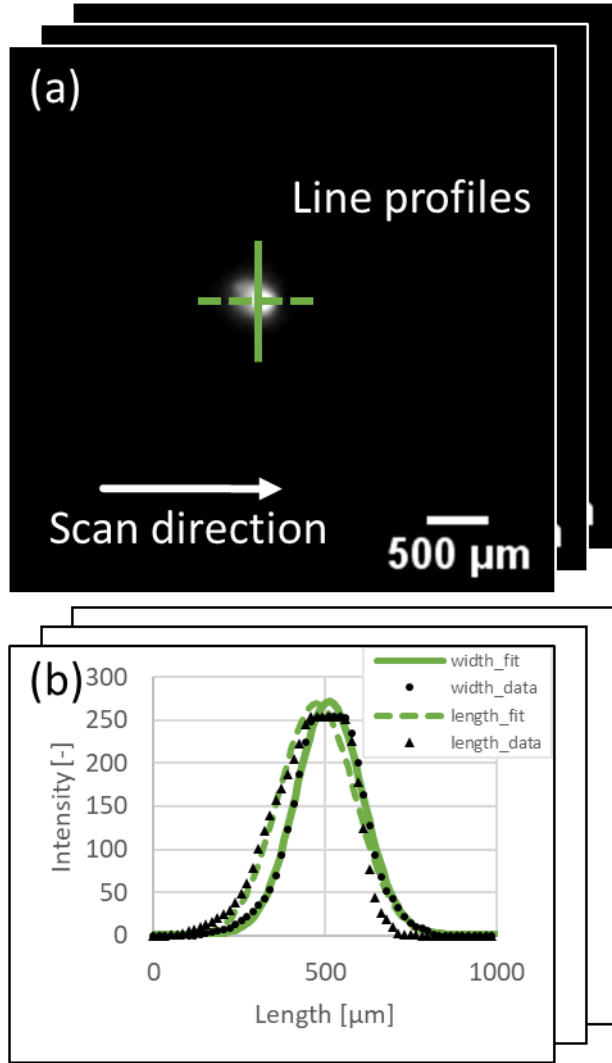


Figure A2: Representative process of melt pool monitoring illustrated with a single track printed at 375 W and 100 mm/s laser parameter. (a) Line profiles along and across melt pool are extracted on all frames recorded when laser is on. (b) Line profiles are fitted using Gaussian function and full width at the half maximum is used to monitor melt pool size variation.

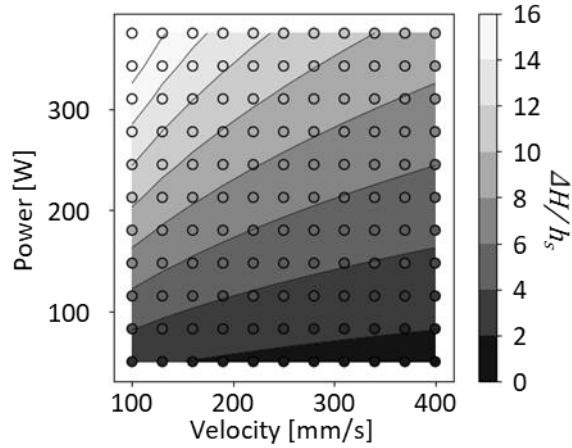


Figure A3: Power – velocity map of the normalized enthalpy calculated following the model developed by Hann *et al.*, 2011 [44]. Normalized enthalpy is low at high-speed – low power and increases with increasing power and decreasing velocity.

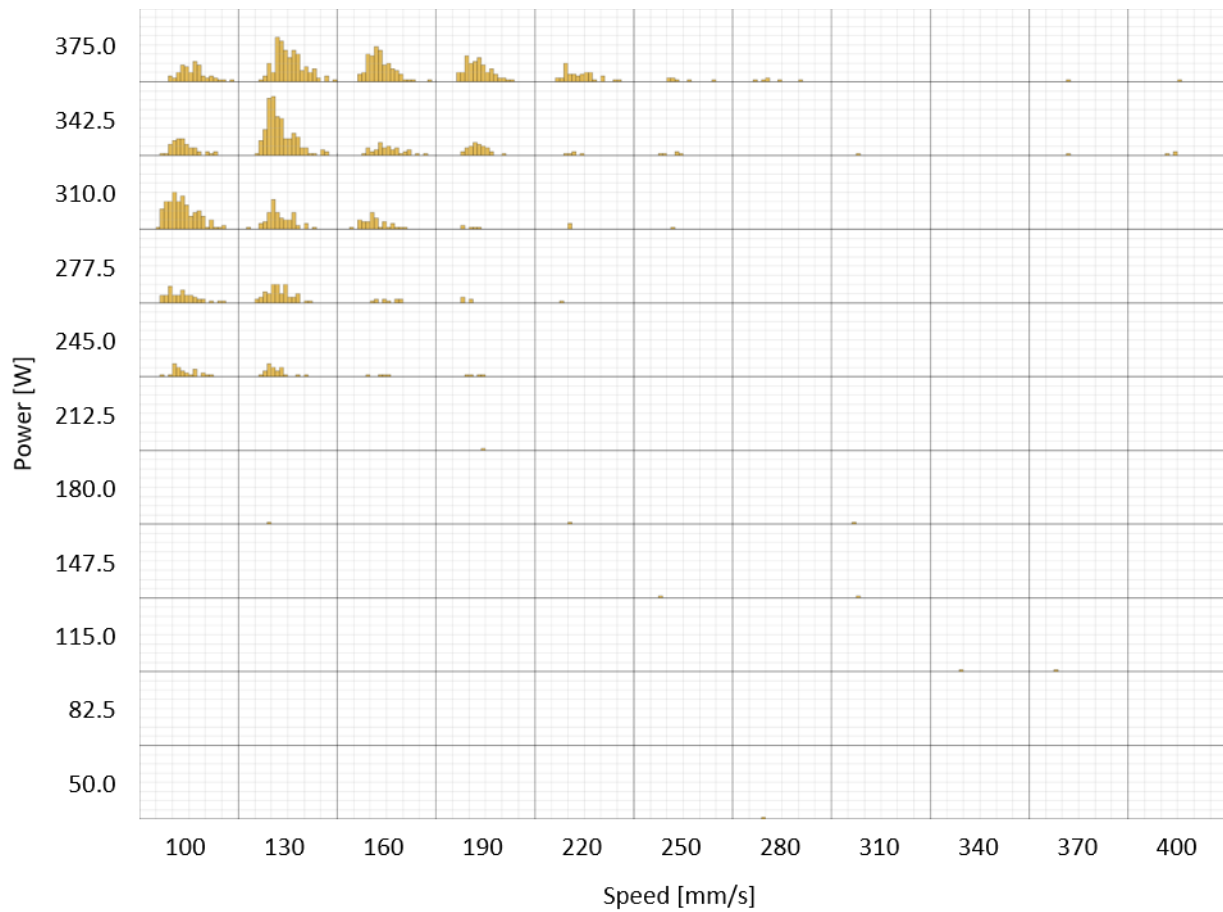


Figure A4: Power-velocity map of pores diameter distributions for each laser power and velocity. Pore counts and pore diameter in μm were evaluated over ranges [0, 40] and [0, 120] respectively.

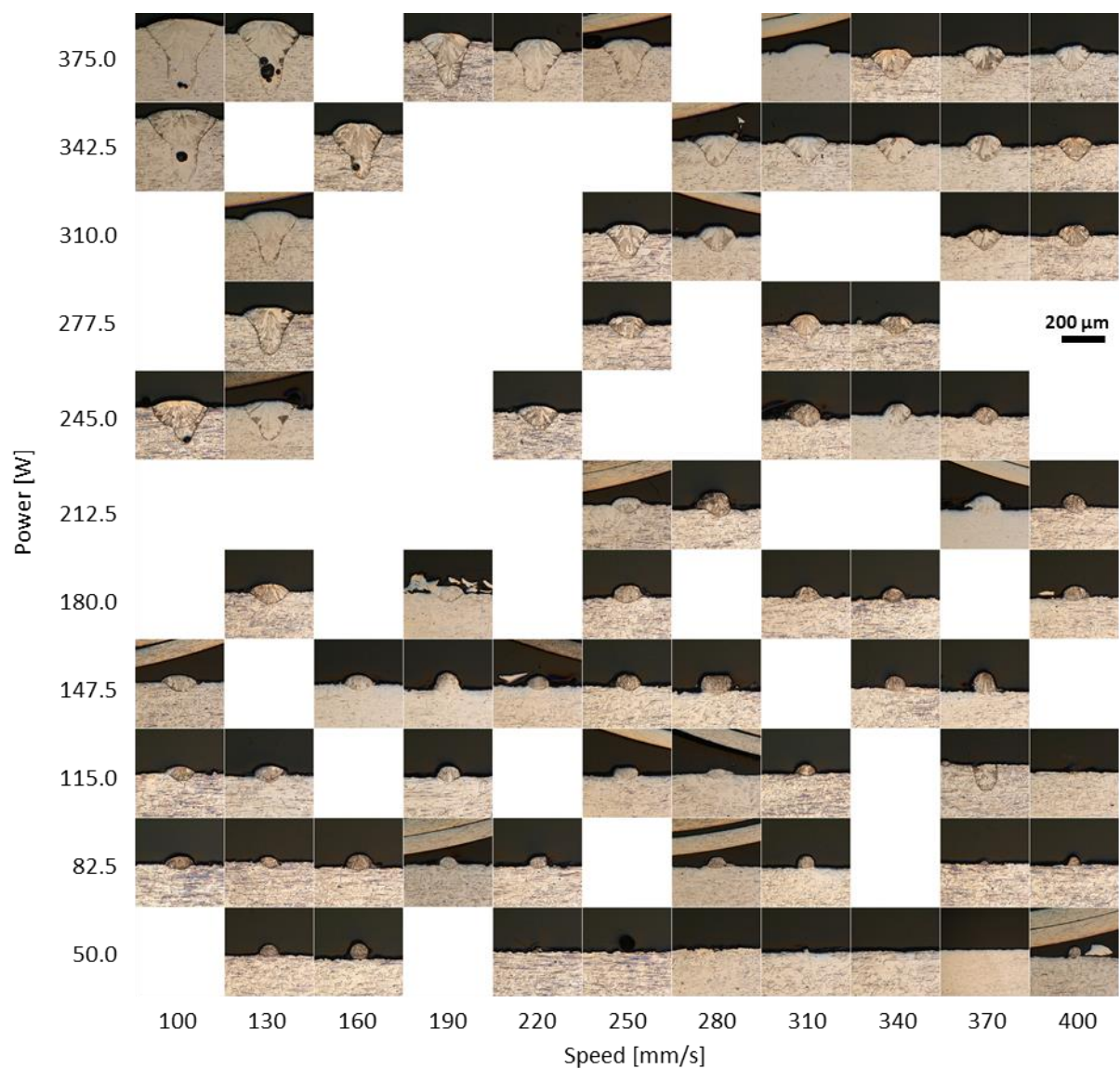


Figure A5: Power – velocity map of metallography cross-sections showing the width and depth of each melt pool. Note that only a few samples were carefully selected for metallography.

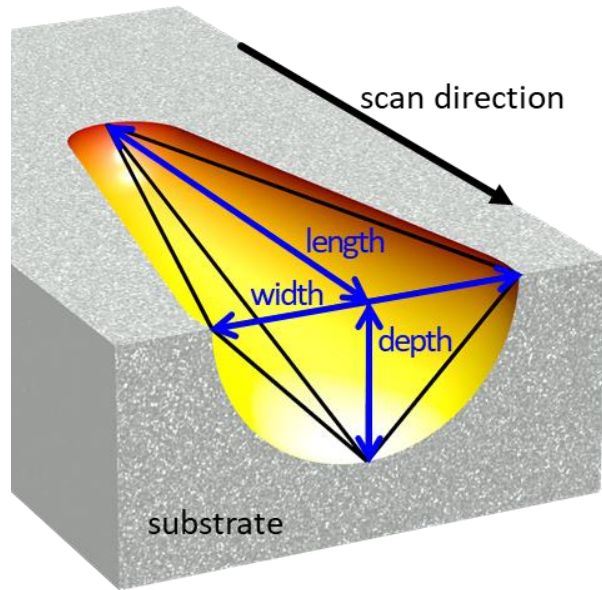


Figure A6: Schematic representation of simplified melt pool geometry used to quantify its volume at different laser power and velocity. The melt pool depth, width and length can vary from 0 to 500 μm depending on the deposited energy.

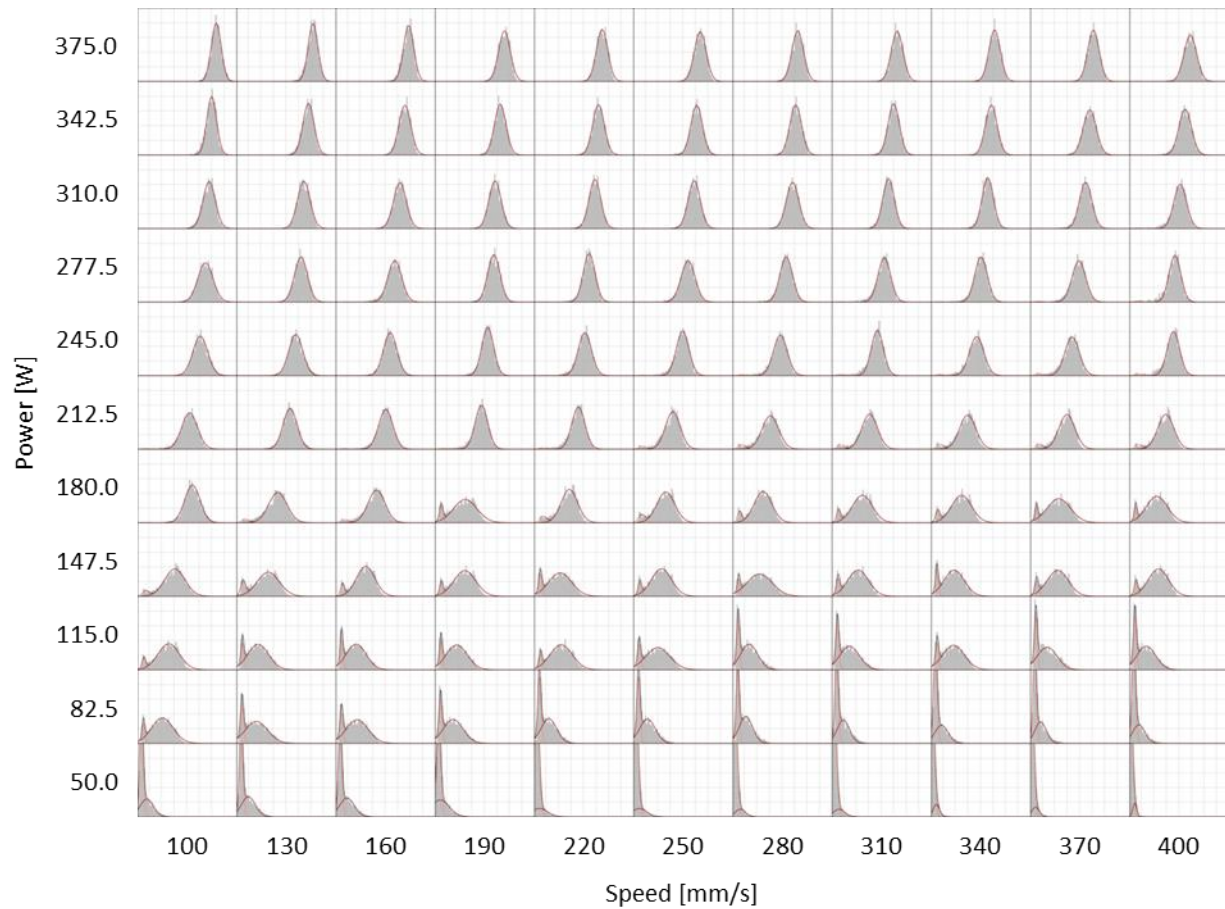


Figure A7: Power-velocity map of the pyrometry histograms. At low powers, the histograms show a sharp peak attributed to low melt pool temperature and conduction mode heating. With increasing power and reducing velocity, a second peak appears, which is attributed to warmer melt pool temperatures. Peaks were fitted using a double gaussian function. The center of the first gaussian peak was constrained to not exceed a pyrometry value higher than 40, and the center of the second gaussian distribution was constrained to not be lower than 25.